Mark

Student's names

Signature

Al-Kufa University

Engineering College

Electronic And Communication Department

3rd stage

LABORATORY MANUAL

Sequential Logic Circuit Lab

"Laboratory Experiments of Digital Logic Circuits"

Experiment 7

Traffic Light Controller System

Case study 3

Part1

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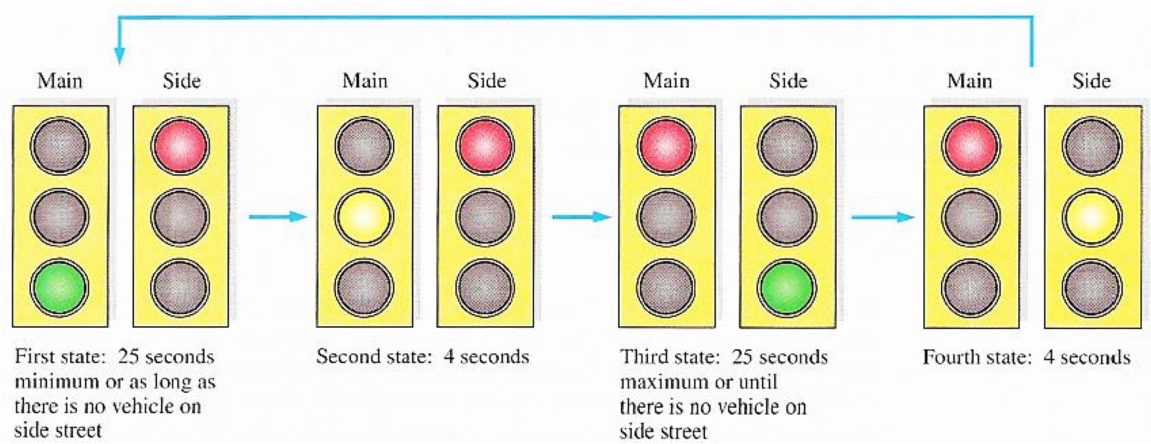
1. Aims:

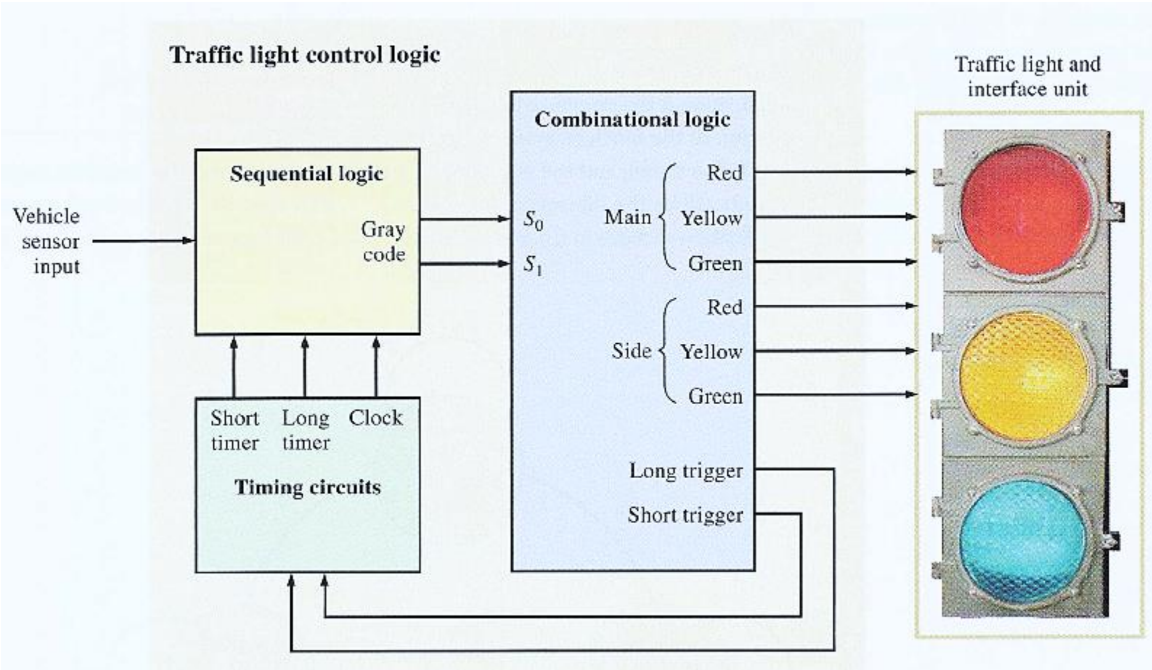
- To understand how to set up a 555 timer as a one-shot.
- To design and implement simple traffic light controller project as an example of digital systems.

2. Theory :

In this experiment a digital system that control a traffic light at the intersection of a busy main street and an occasionally used side street is going to be designed. The priority for cars to pass will be of course for the main street where the green light will be at least 10s on or as long as there is no vehicle on the side street. The side street will have a green light until there is no vehicle on the side street or for maximum of 10 s. In the design, there must be a caution light (yellow) for 3s between changes from green to red on both main and side streets. These requirement are illustrated in the diagram below.



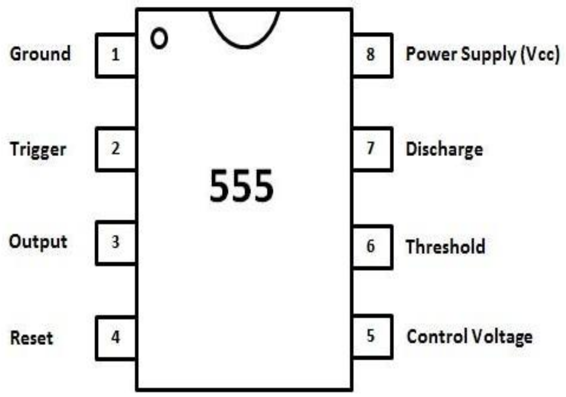
From the requirements, you can develop a block diagram of the system. First, you know that the system must control six different pairs of lights. These are the red, yellow, and green lights for both directions on the main street and the red, yellow, and green lights for both directions on the side street. Also, you know that there is one external input (other than power) from a side street vehicle sensor. The figure below is a minimal block diagram showing these requirements.

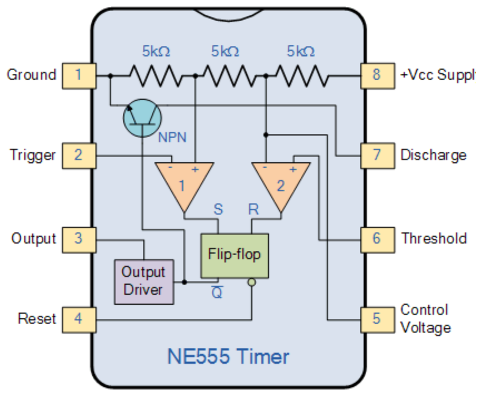


In this experiment the timing circuit will be designed. The timing circuit consists of two parts 4s and 10s timers as well as a clock generator. The 4s and 10s timers are implemented using 555 IC and the clock will be taken from the signal generator device.

2.1 The 555 Timer

The 555 timer is a versatile and widely used IC device because it can be configured in two different modes as either a monostable multivibrator (one-shot) or as an astable multivibrator (oscillator). An astable multivibrator has no stable states and therefore changes back and forth (oscillates) between two unstable states without any external triggering. While monostable multivibrator needs an external trigger.



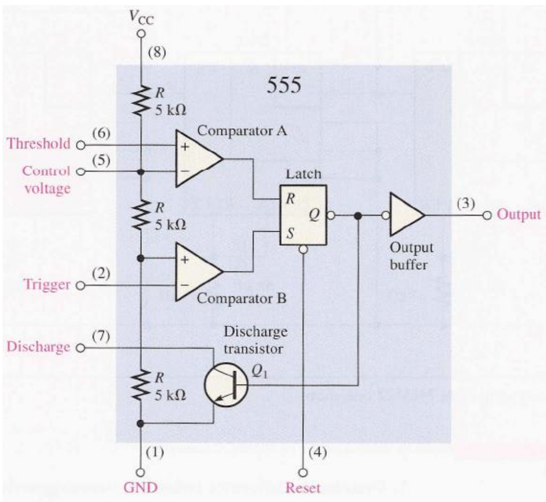


2.1.1 Basic Operation

The **555 Timers** name comes from the fact that there are three 5kΩ resistors connected together internally producing a voltage divider network between the supply voltage at pin 8 and ground at pin 1. The voltage across this series resistive network holds the negative inverting input of comparator two at 2/3Vcc and the positive non-inverting input to comparator one at 1/3Vcc.

A functional diagram below showing the internal components of a 555 timer. The comparators are devices whose outputs are HIGH when the voltage on the positive (+) input is greater than the voltage on the negative (-) input and LOW when the – input voltage is greater than the + input voltage. The voltage divider consisting of three 5 k resistors provides a trigger level of 1/3Vcc and a threshold level of 2/3Vcc. The control voltage input (pin 5) can be used to externally adjust the trigger and threshold levels to other values if necessary. When the normally HIGH trigger input momentarily goes below 1/3Vcc.

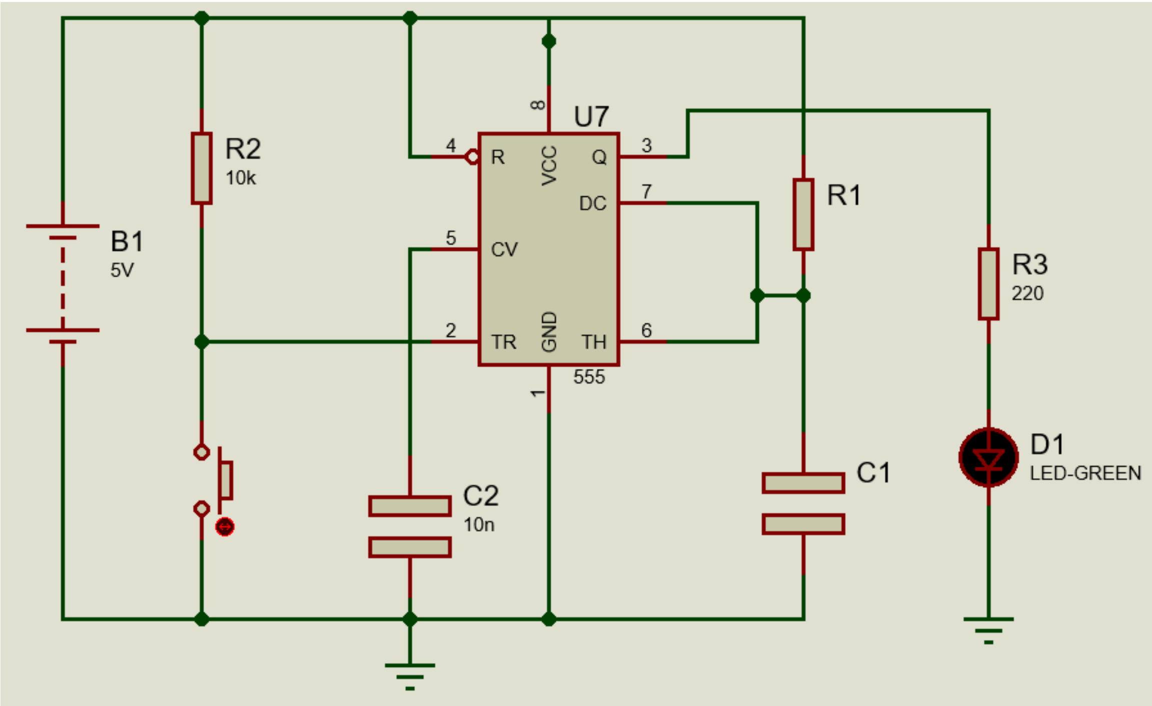
the output of comparator B switches from LOW to HIGH and sets the S-R latch, causing the output (pin 3) to go HIGH and turning the discharge transistor Q, off. The output will stay HIGH until the normally LOW threshold input goes above 2/3Vcc and causes the output of comparator A to switch from LOW to HIGH. This resets the latch, causing the output to go back LOW and turning the discharge transistor on. The external reset input can be used to reset the latch independent of the threshold circuit. The trigger and threshold inputs (pins 2 and 6) are controlled by external components connected to produce either monostable or astable action.



2.1.2 Monostable (one shot) operation

An external resistor R1 and capacitor C1 connected as shown in the Figure below are used to set up the 555 timer as a one-shot. The pulse width of the output is determined by the time constant of R_1 and C_1 according to the following formula:

$$t_W = 1.1 R_1 C_1$$



3. Procedure:

1. From the block diagram in section 2.1.2, design and connect a 10s timer by choosing suitable values for R1 and C1.

2. Replace the switch button in your design by an electronic triggering circuit. The trigger circuit should be able to convert a constant 5v into a one short pulse.

3. Draw your full circuit design with details.